



TEKNOLOGI INFORMASI SISTEM ADMINISTRASI JARINGAN

IT NETWORK SYSTEMS ADMINISTRATION

Modul – Network Challenge (Cisco)

Actual Task

**LOMBA KOMPETENSI SISWA SEKOLAH MENENGAH KEJURUAN
TINGKAT DKI Jakarta Wilayah Jakarta Timur Tahun 2024**

Module Network Challenge (Cisco)

Description of project and tasks

This test project is designed using a variety of network technologies that should be familiar from the Cisco certification tracks. Tasks are broken down into following configuration sections:

- Basic configuration
- Switching
- Routing
- Services
- Security
- WAN

All sections are independent but all together they build very complex network infrastructure. Some tasks are pretty simple and straightforward; others may be tricky. You may see that some technologies are expected to work on top of other technologies. It is important to understand that if you are unable to come up with a solution in the middle of such a technology stack it doesn't mean that the rest of your work will not be graded at all. For example, you may not configure IPv4 routing that is required for VPN because of IP reachability, but you can use static routes and then continue to work with VPN configuration and everything that runs on top. You won't receive points for IPv4 routing in this case, but you will receive points for everything that you made operational on top as long as functional testing is successful.

Instructions Notice to the Competitor

Your configuration will be marked with scripts, so therefore we need two important basic configurations:

1. no ip domain-lookup
2. exec-timeout 0 0 on console

Both configurations are already pre-configured on all switches and routers, so do not change these configurations.

Instructions to the Competitor

1. Read all tasks in each section before proceeding with any configuration. The completion of any item may require the completion of any previous or later item.
2. Points are awarded for working configurations only. Test the functionality of all the requirements, before you submit the test project. Be careful, because as you configure one part, you may break a previous requirement or configuration.
3. No partial points can be granted for any aspect; all requirements need to be fulfilled to receive the points for the aspect. Some requirements depend on other aspect's requirements, either before or after the current aspect.
4. Save your configurations frequently; accidents do and will happen.
5. Hosts are preconfigured but check the configuration and change it when necessary.
6. Please use industrial best practice where possible!

Basic configuration

1. Configure hostnames for all network devices as you see on the topology.
2. Configure domain name **lks2024.id** for all network devices on the topology.
3. Configure **Skill39** as a privileged mode password for all devices.
4. Configure IPv4/IPv6 address for all network devices as you see on the topology.
5. Configure GMT +7 as a timezone for all network devices.

Switching

1. Configure VTP on all switches to synchronize VLANs. It should be possible to modify VLAN database only from L3SW-1 with VTP version 1, and VLAN databases of all the other switches should be synchronized from L3SW-1. VLAN database on all switches should contain following VLANs.
 - a. VLAN 10 with name SRV
 - b. VLAN 20 with name CLI
2. Configure all links between switches as trunk port.
 - a. Do not use dynamic negotiation protocol.
 - b. Configure manual pruning so that only created VLANs are allowed forwarding.
3. Configure EtherChannel between switches.
 - a. Use following port-channel numbers:
 - i. 1 - between switches L3SW-1 and L3SW-2
 - ii. 2 - between switches L3SW-1 and L2SW-1
 - iii. 3 - between switches L3SW-2 and L2SW-2
 - b. The aggregated channel between L3SW-1 and L3SW-2 do not use dynamic negotiation protocol.
 - c. The aggregated channel between L3SW-1 and L2SW-1 use a Cisco proprietary protocol for dynamic negotiation.
 - d. The aggregated channel between L3SW-2 and L2SW-2 use a standard protocol for dynamic negotiation.
 - e. L3SW-1 and L3SW-2 should initiate negotiation and the other devices should respond but don't initiate.
 - f. Configure the load balancing and forwarding method with source and destination MAC address.
4. Spanning tree configuration.
 - a. L3SW-1 should be root bridge of VLAN10. If L3SW-1 goes down L3SW-2 should take over as the root bridge.
 - b. L3SW-2 should be root bridge of VLAN20. If L3SW-2 goes down L3SW-1 should take over as the root bridge.
 - c. The traffic from ES-CLI should pass through L3SW-2.
 - d. Configure port which is connected to end device so that it immediately begins forwarding when connected.

Routing

1. Configure FHRP on L3SW-1 and L3SW-2.
 - a. Use Hot Standby Router Protocol v2 for VLAN 10.

- i. L3SW-1 should be used as the default gateway and use HSRP priority 110 for this switch
 - ii. Use 104 as the group number of IPv4.
 - iii. Use 192.168.10.1 as virtual IPv4.
 - iv. Standby switch should immediately take over when active router down
 - b. Use a Hot Standby Router Protocol v2 for VLAN 20.
 - i. L3SW-2 should be used as the default gateway and use HSRP priority 110 for this switch.
 - ii. Use 204 as the group number of IPv4.
 - iii. Use 192.168.20.1 as virtual IPv4 address.
 - iv. Standby switch should immediately take over when active router down
2. Configure default route to ISP on both EST-1 and EST-2.
3. Configure EIGRP between EST router and L3SW
 - a. use as number 39
 - b. make sure network vlan 10 and 20 can be reach EST-1
 - c. Redistribute default route from EST-1 and EST-2 into EIGRP
4. Configure OSPF between ISP router, EST and WST router.
 - a. Use OSPF area 0 and instance number 11.
 - b. Advertise only Public IP on all routers.
 - c. Configure OSPF so that routing updates are not sent into networks where they are not required.

Services

1. Configure DHCP
 - a. DHCP server must be configure on L3SW-1 with pool name ESCLI
 - b. Exclude IP address from range 192.168.20.1 until 192.168.20.10
 - c. ES-CLI can obtain IP address automatically.
 - d. All DHCP client should use WS-SRV as DNS Server.
2. Configure NAT
 - a. Configure NAT on EST-1 and EST-1 with NAT Pool name EST_Pool and include all ES endpoint or server into that pool
 - b. When ES-CLI communicate with internet, these IP address should be translated to 103.76.14.1 - 103.76.14.10 (subnet /28)
3. Configure web server on ES-SRV
 - a. Enable http and https services on ES-SRV
 - b. Ensure web server on ES-SRV can be reach using domain jt2.lks2024.id
4. Configure ISP as NTP server. All Router devices should synchronize time from ISP loopback0 IP address.

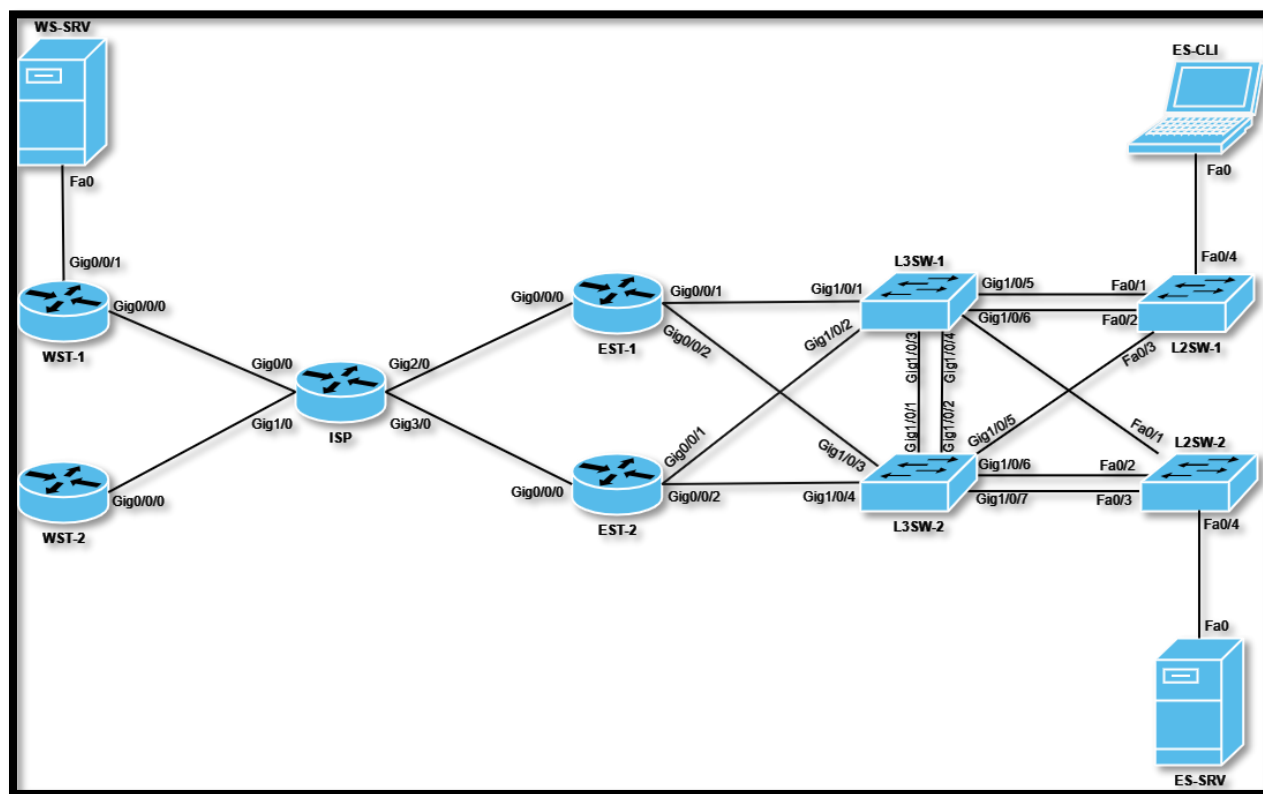
Security

1. Configure SSH version 2 for remote access on EST-1 and EST-2 with below username and password **Skill39** :
 - username user1 with maximum priviledge level
 - username user2 with priviledge level 5
 - a. User user2 should be able to configure any interface IP settings and administratively enable or disable any of these interfaces.
 - b. Ensure only ES-CLI is allowed to access via SSH to EST-1 and EST-2

2. Configure port-security on the port which is connected to ES-CLI using following parameters:
 - a. Maximum MAC address – 2
 - b. In case of policy violation, security message should be displayed on the console, port should be disabled.

Appendix

Topology



IP Address Configuration Table

Site	Device	Interface	IP Address
WEST	WST-1	Gig0/0/0	103.255.10.2/24
		Gig0/0/1	139.255.10.1/24

	WST-2	Gig0/0/0	103.255.20.2/24
	WS-SRV	Fa0	139.255.10.51/24
Internet	ISP	Gig0/0	103.255.10.1/24
		Gig1/0	103.255.20.1/24
		Gig2/0	103.255.30.1/24
		Gig3/0	103.255.40.1/24
		Loopback0	1.1.1.1/32
EAST	EST-1	Gig0/0/0	103.255.30.2/24
		Gig0/0/1	172.16.1.1/30
		Gig0/0/2	172.16.1.5/30
	EST-2	Gig0/0/0	103.255.40.2/24
		Gig0/0/1	172.16.1.9/30
		Gig0/0/2	172.16.1.13/30
	L3SW-1	Gig1/0/1	172.16.1.2/30
		Gig1/0/2	172.16.1.10/30
		Vlan10	192.168.10.2/24
		Vlan20	192.168.20.2/24
	L3SW-2	Gig1/0/3	172.16.1.6/30
		Gig1/0/4	172.16.1.14/30
		Vlan10	192.168.10.3/24
		Vlan20	192.168.20.3/24
	ES-CLI	Fa0	192.168.20.x/24 (DHCP)
	ES-SRV	Fa0	192.168.10.254/24